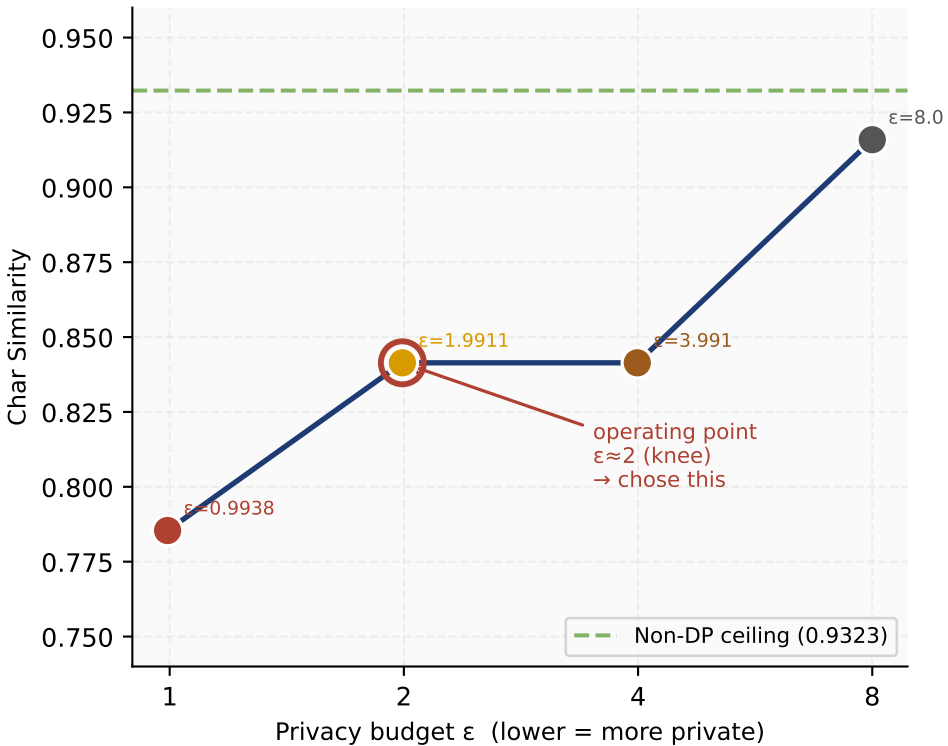


Script 5: Hyperparameter Justification Dashboard

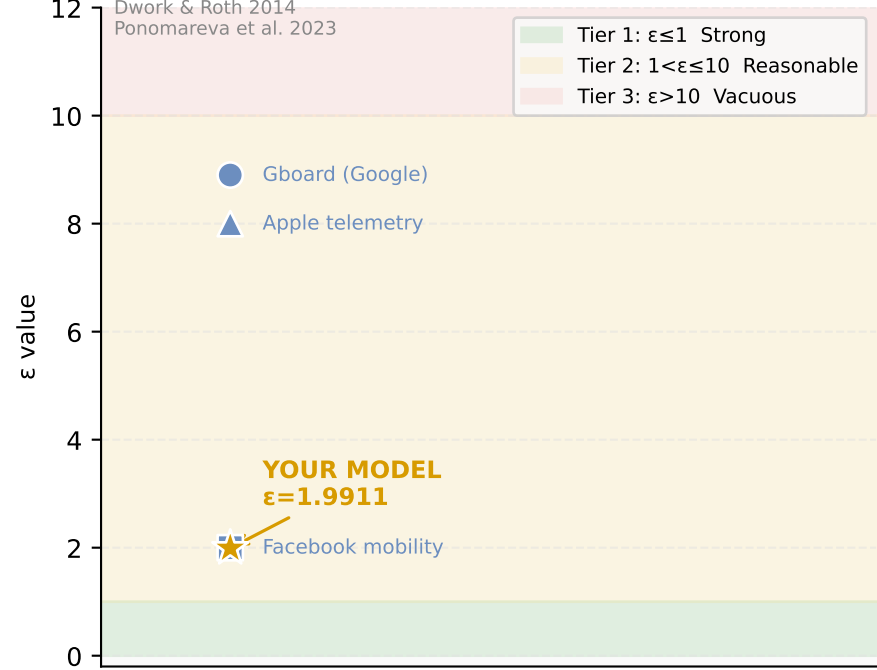
DP-QLoRA Payee Extraction — Why Each Parameter Was Chosen

$\epsilon=1.9911 \cdot \delta=2.5 \times 10^{-4} \cdot C=1.5 \cdot B=24 \cdot r=16 \cdot 6 \text{ epochs} \cdot \text{MAX_LENGTH}=64$

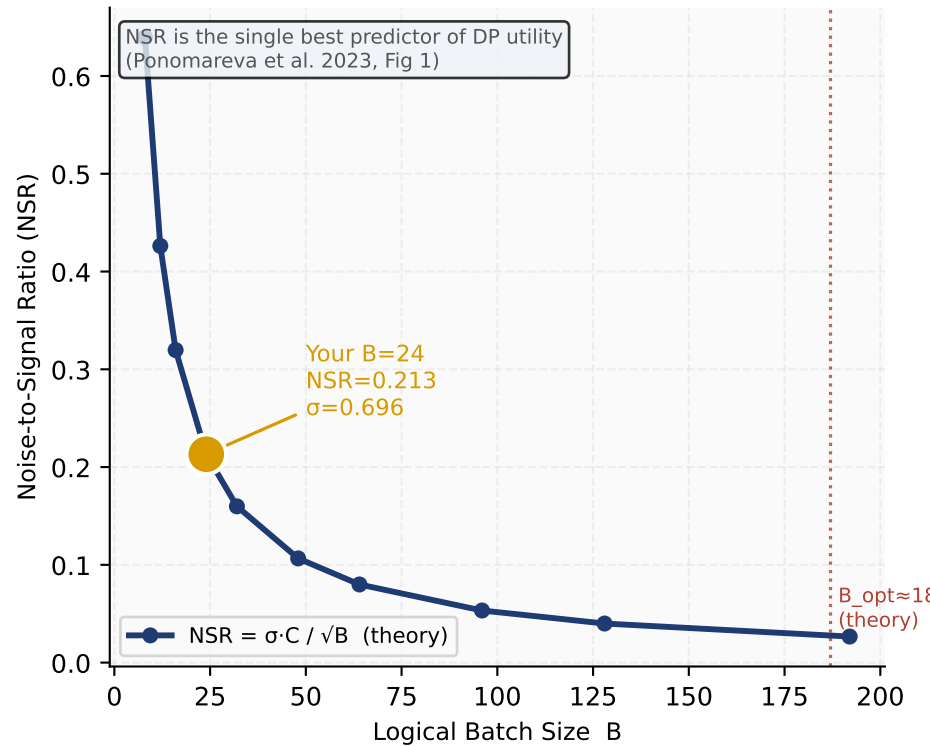
A: ϵ Sweep — Why $\epsilon=2$?
Knee: largest utility, minimal privacy cost



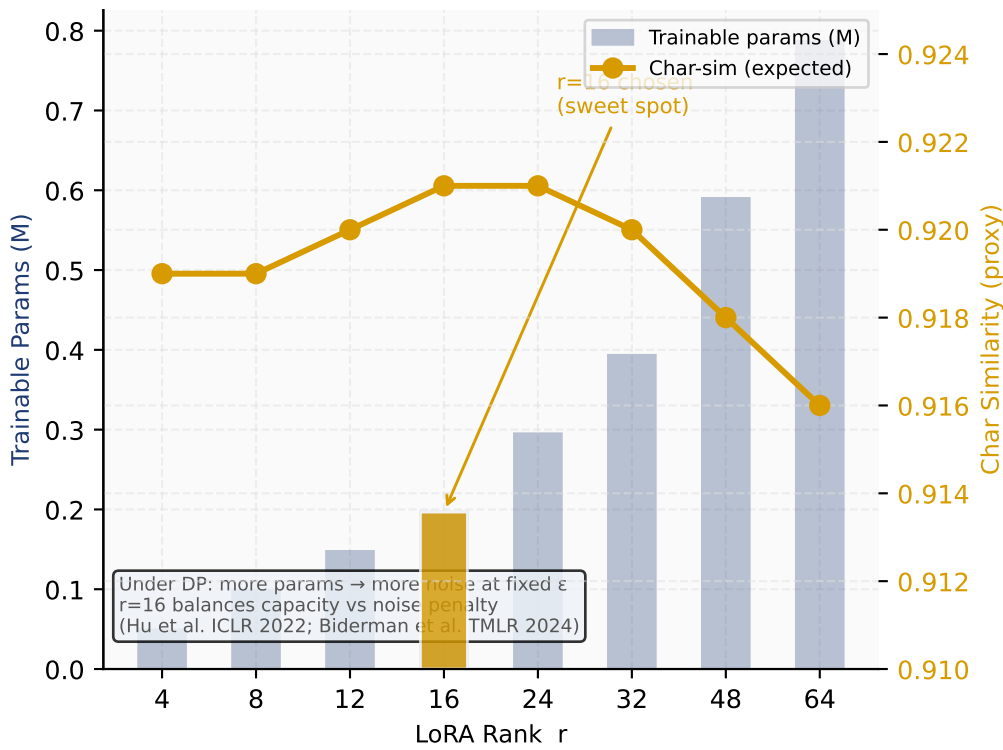
B: Privacy Tier Framework
Your model vs production deployments



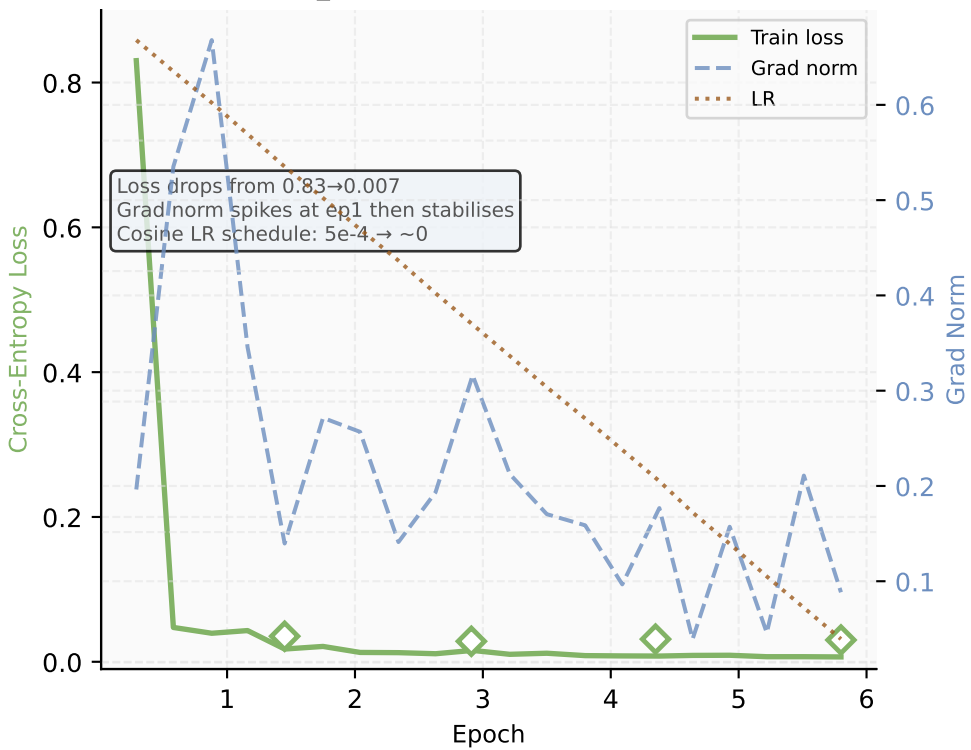
C: NSR vs Batch Size
Larger batch = less noise at same ϵ



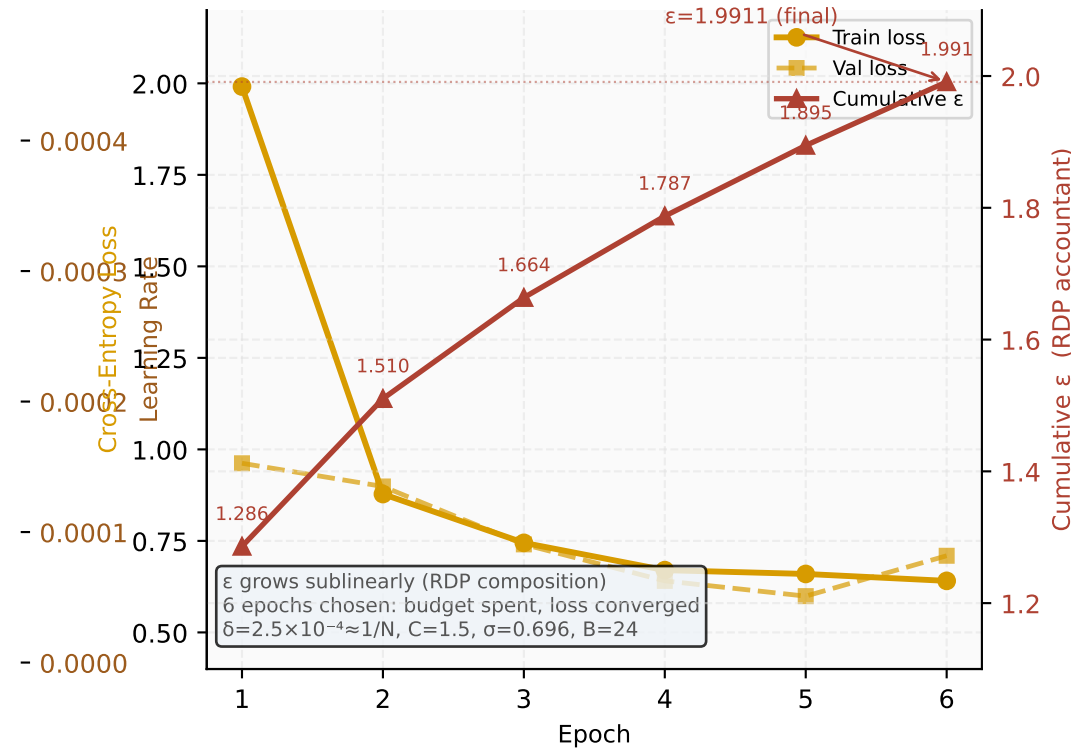
D: LoRA Rank $r=16$
Capacity vs DP noise tradeoff



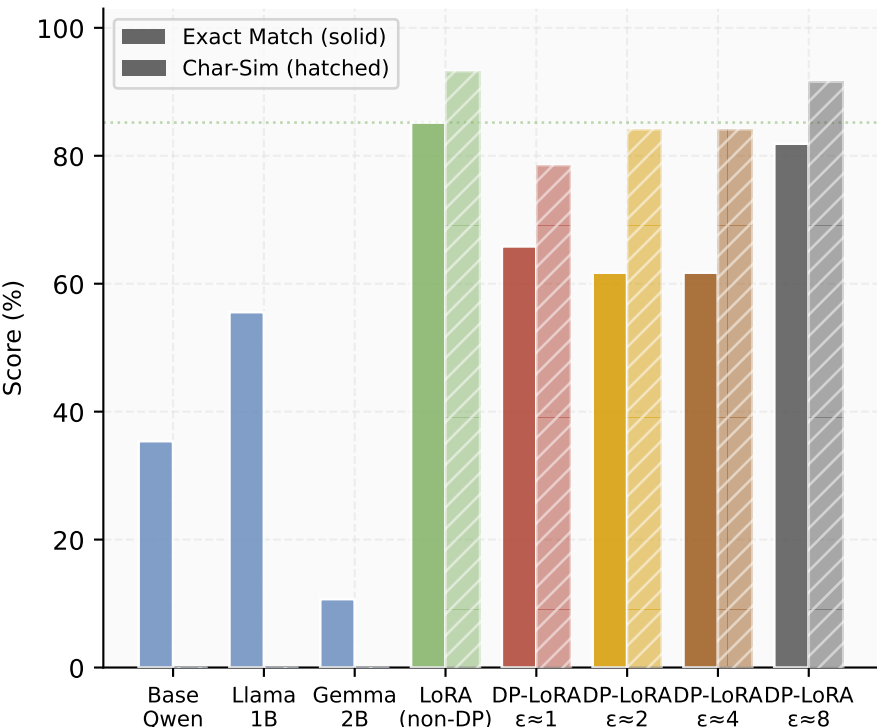
E: Non-DP QLoRA Training Dynamics
Real trainer_state (checkpoint-400, 414 steps)



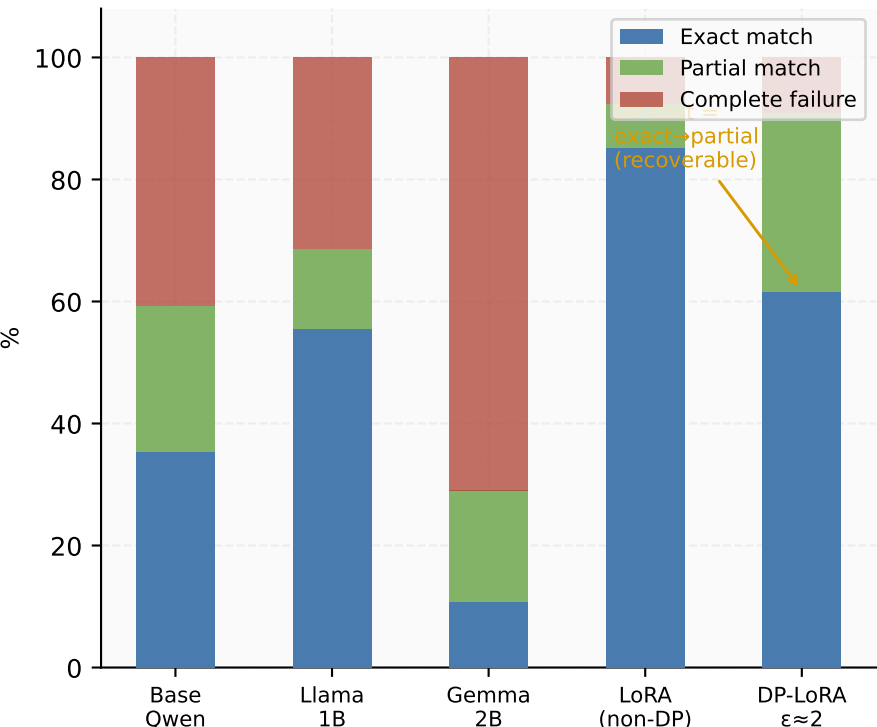
F: DP-QLoRA Training Dynamics ($\epsilon \approx 2$)
RDP accountant tracks ϵ epoch-by-epoch



G: Full Model Comparison
EM (solid) vs Char-Sim (hatched)



H: DP Cost Anatomy
Exact to Partial shift, not catastrophic failure



I: NSR Across All Four ϵ Runs
Lower ϵ to more noise to higher NSR

